

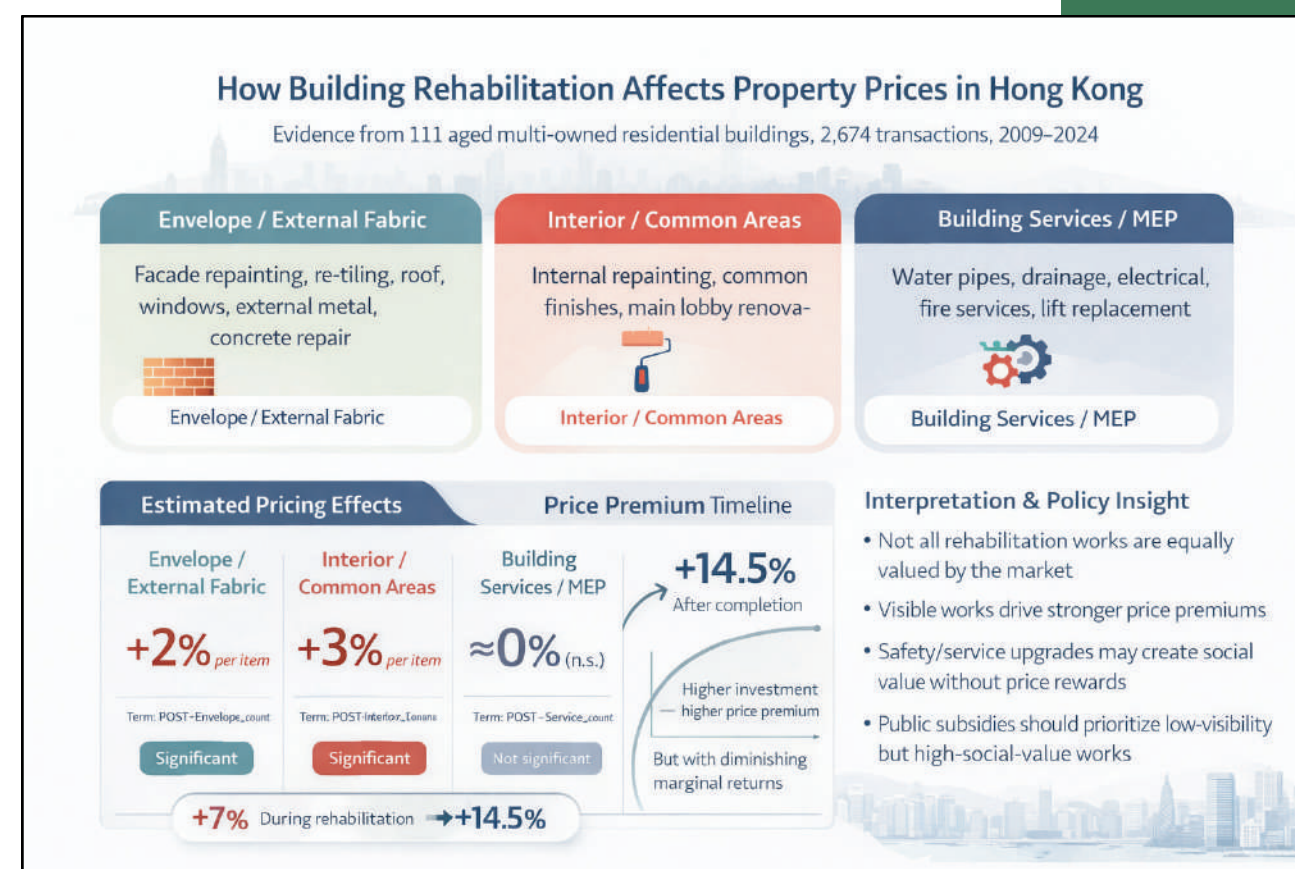
BSc(Surveying) | RECO40II Dissertation

Does Major Building Rehabilitation Affect Residential Property Price? A Hedonic Regression Analysis on Work-Scope Categories (Hong Kong, 2009–2024)

STUDENT: Ko Hui Ki

SUPERVISOR: Prof. Frank Xue

Major Findings



1. Property transactions during the rehabilitation process experience a 7% price increase, which rises to 14.5% upon project completion.
2. While higher rehabilitation expenditure per unit of floor area increases property values, it exhibits diminishing marginal returns at higher investment levels.
3. Transaction prices reflect a significant premium for visible improvements to a building's external fabric and common areas, whereas concealed building services do not yield statistically significant returns.

Abstract

In Hong Kong, a rapidly ageing stock of multi-owned private residential buildings makes urban decay a pressing issue. While redevelopment is often proposed, it is frequently constrained by fragmented ownership and limited economic viability. Consequently, building rehabilitation has emerged as the primary urban renewal strategy. This dissertation addresses the research gap regarding how specific dimensions of rehabilitation, including investment scale and work categories, influence property prices.

To achieve this aim, the study examined 111 rehabilitated aged stand-alone buildings in Kowloon City and Sham Shui Po, completed between 2009 and 2024. Utilising 2,674 transaction records within a seven-year event window, the research employs three hedonic pricing models to test:

1. The existence of price premiums during and after rehabilitation.
2. The impact of monetary investment scale and potential diminishing marginal returns.
3. The valuation of different work categories: "External Fabric," "Common Areas," and "Building Services/MEP."

The empirical results support the hypotheses. Transactions during rehabilitation exhibit a 7 per cent premium, which increases to 14.5 per cent upon project completion. Additionally, improvements to the external envelope and interior or common areas are significantly valued by the market, whereas "Building services or MEP" work, does not have a significant impact on transaction prices. Higher investment per unit floor area is associated with greater property value, although the relationship demonstrates diminishing marginal returns. These findings suggest that subsidy frameworks could be differentiated, as the secondary market capitalises on rehabilitation primarily through visible aesthetic enhancements while largely ignoring hidden risk-reduction benefits. Public resources may be more effectively directed toward social-value works, such as critical safety and service upgrades that yield weak price effects, whereas market-driven aesthetic improvements can rely on private incentives. This provides a conceptual roadmap for allocating public funds to sustain the city's ageing private housing stock.

